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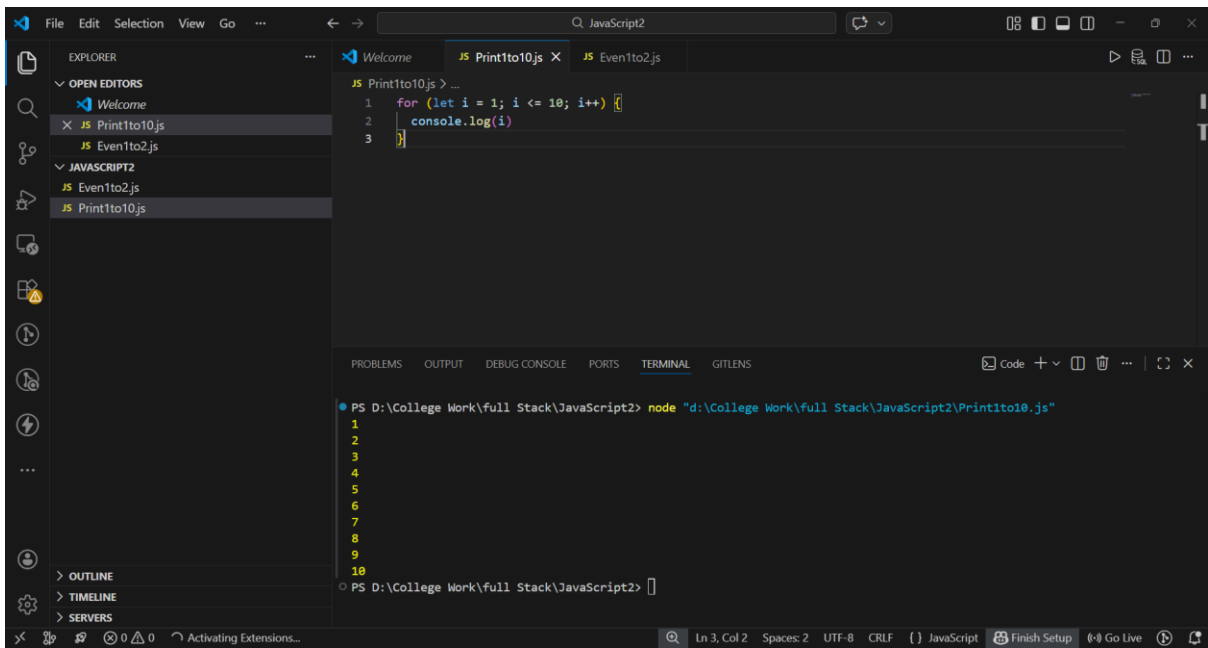
Class: MCA-1 Div: A

Roll No: 08

Subject: Modern Full Stack

For Loop Tasks :

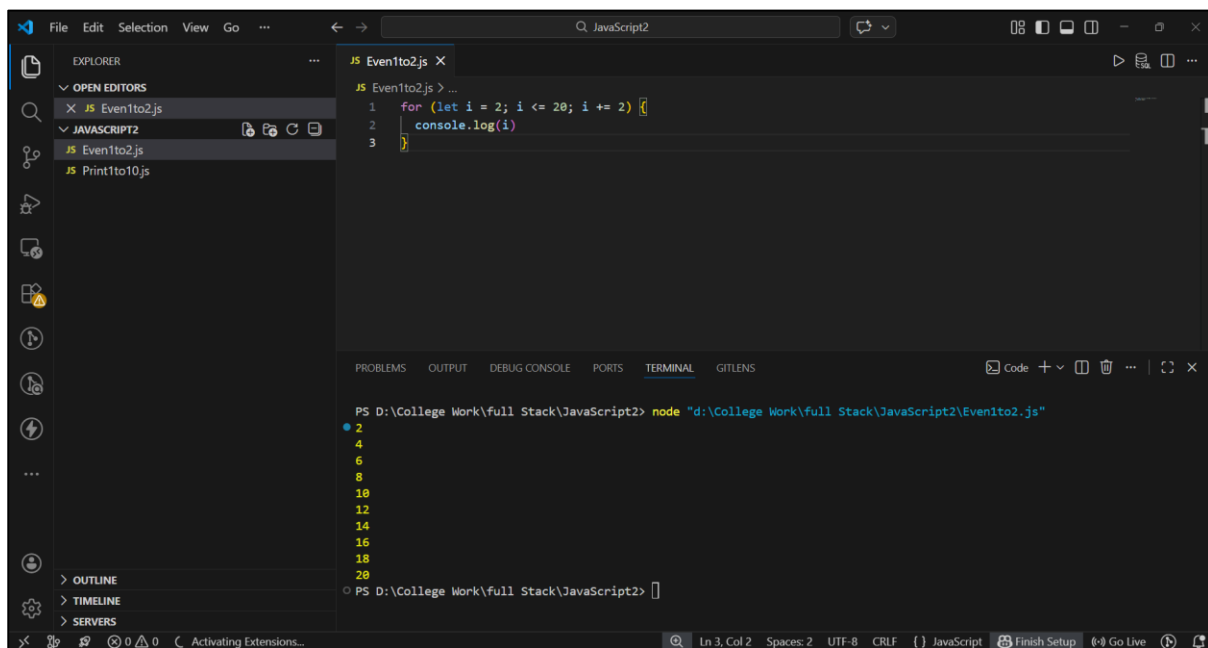
1. Print numbers from 1 to 10. (Using For Loop Task)



```
JS Print1to10.js > ...
1 for (let i = 1; i <= 10; i++) {
2   console.log(i)
3 }
```

```
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Print1to10.js"
1
2
3
4
5
6
7
8
9
10
PS D:\College Work\full Stack\JavaScript2>
```

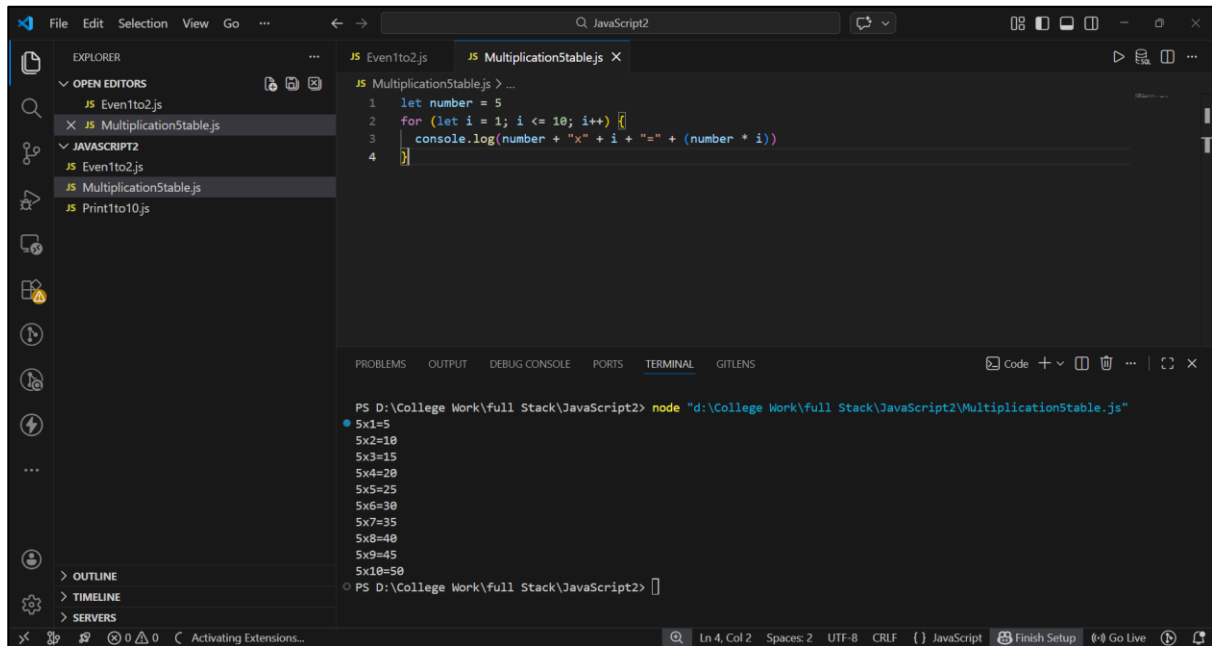
2. Print all even numbers between 1 and 20.



```
JS Even1to2.js > ...
1 for (let i = 2; i <= 20; i += 2) {
2   console.log(i)
3 }
```

```
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Even1to2.js"
2
4
6
8
10
12
14
16
18
20
PS D:\College Work\full Stack\JavaScript2>
```

3. Take a number (e.g., 5) and print its multiplication table up to 10 (5 x 1 = 5... 5 x 10 = 50).

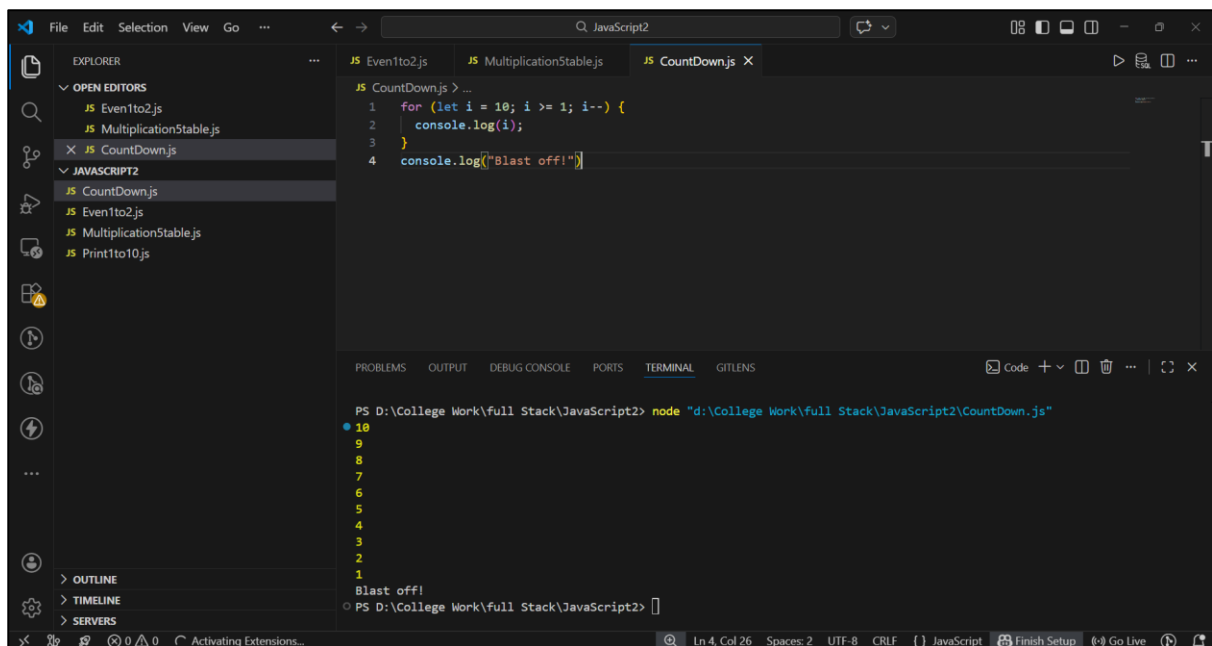


```
JS MultiplicationTable.js > ...
1 let number = 5
2 for (let i = 1; i <= 10; i++) {
3   console.log(number + "x" + i + "=" + (number * i))
4 }
```

PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\MultiplicationTable.js"

```
5x1=5
5x2=10
5x3=15
5x4=20
5x5=25
5x6=30
5x7=35
5x8=40
5x9=45
5x10=50
PS D:\College Work\full Stack\JavaScript2>
```

4. Start from 10 and count down to 1, then print "Blast off!"



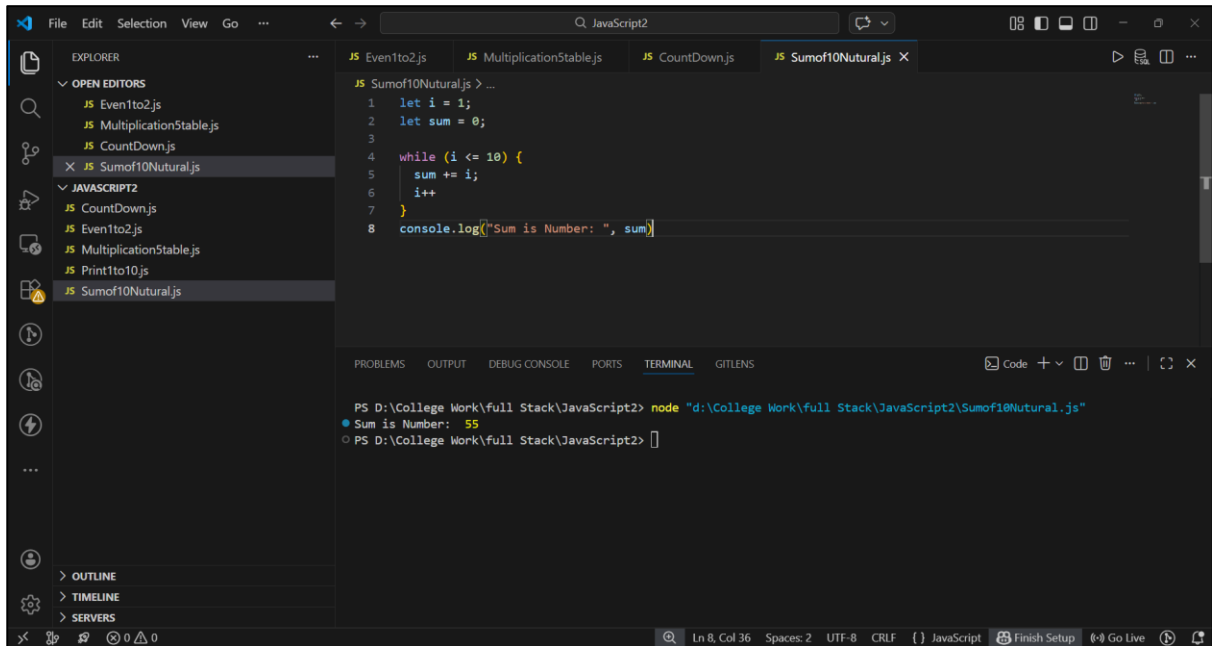
```
JS CountDown.js > ...
1 for (let i = 10; i >= 1; i--) {
2   console.log(i);
3 }
4 console.log("Blast off!")
```

PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\CountDown.js"

```
10
9
8
7
6
5
4
3
2
1
Blast off!
PS D:\College Work\full Stack\JavaScript2>
```

While Loop Task :

1. Find the sum of the first 10 natural numbers (1+2+3...+10)



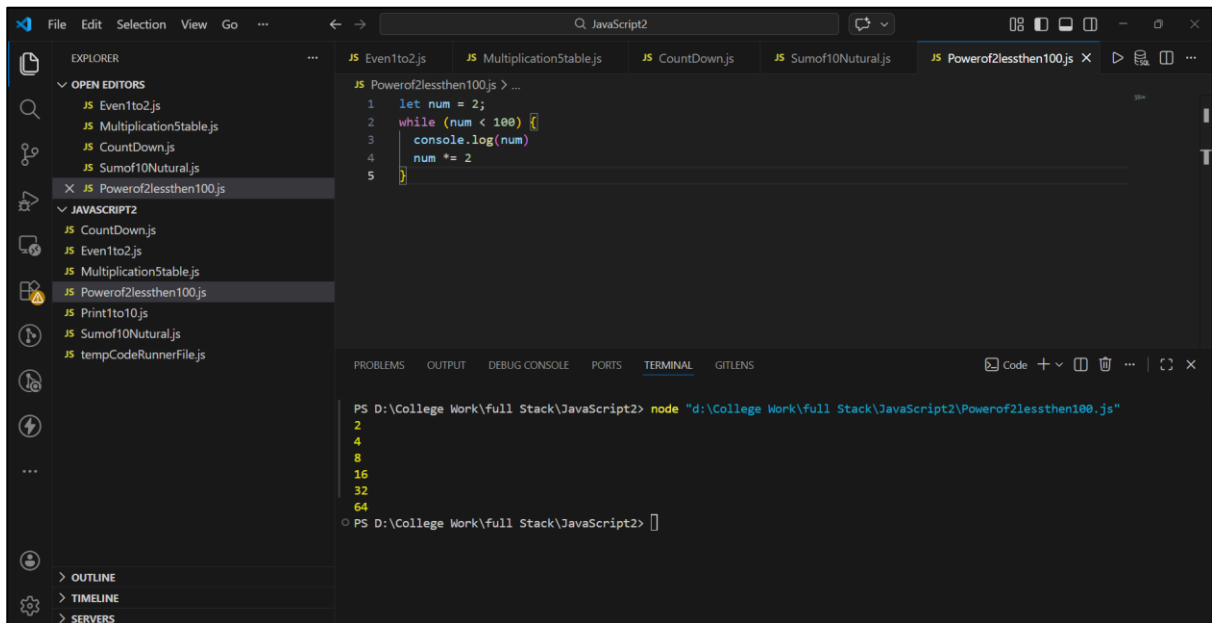
The screenshot shows the Visual Studio Code editor with a file named `Sumof10Natural.js` open. The code uses a `while` loop to calculate the sum of the first 10 natural numbers. The terminal output shows the result: "Sum is Number: 55".

```
1 let i = 1;
2 let sum = 0;
3
4 while (i <= 10) {
5   sum += i;
6   i++;
7 }
8 console.log("Sum is Number: ", sum)
```

Terminal Output:

```
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Sumof10Natural.js"
Sum is Number: 55
PS D:\College Work\full Stack\JavaScript2>
```

2. Print the powers of 2 (2, 4, 8, 16...) that are less than 100.



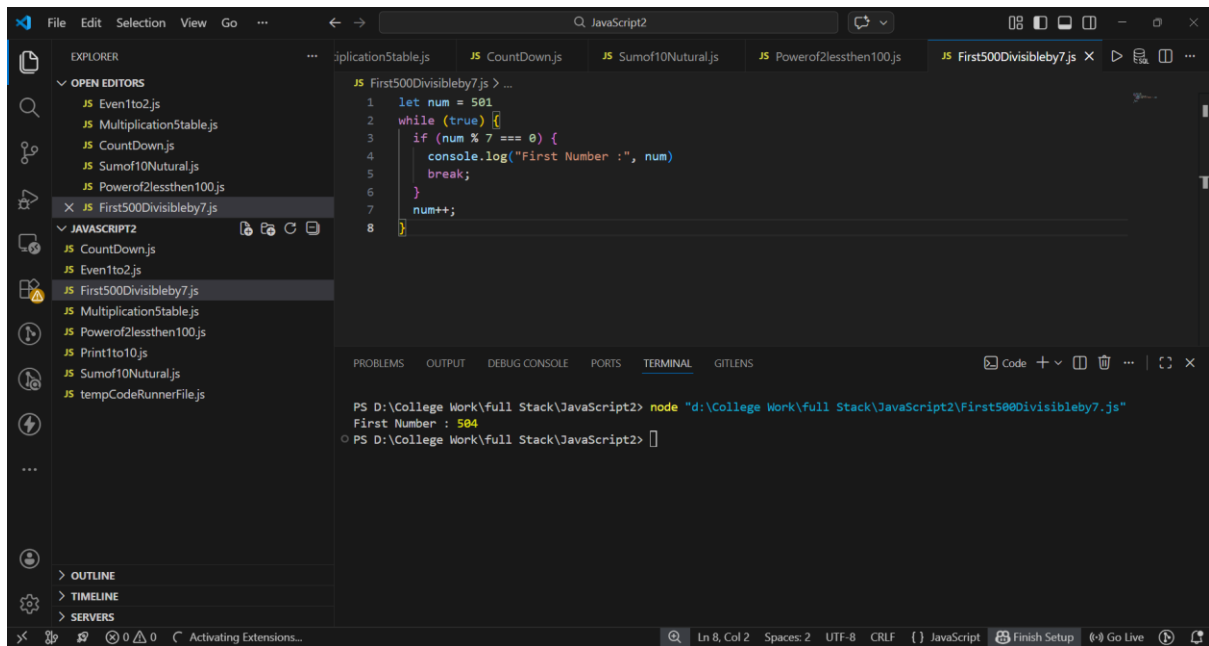
The screenshot shows the Visual Studio Code editor with a file named `Powerof2lessthen100.js` open. The code uses a `while` loop to print powers of 2 that are less than 100. The terminal output shows the sequence: 2, 4, 8, 16, 32, 64.

```
1 let num = 2;
2 while (num < 100) {
3   console.log(num)
4   num *= 2
5 }
```

Terminal Output:

```
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Powerof2lessthen100.js"
2
4
8
16
32
64
PS D:\College Work\full Stack\JavaScript2>
```

3. Use a while loop to find the first number after 500 that is divisible by 7.



The screenshot shows the Visual Studio Code editor with a JavaScript file named `First500Divisibleby7.js` open. The code in the editor is as follows:

```
1 let num = 501
2 while (true) {
3   if (num % 7 === 0) {
4     console.log("First Number :", num)
5     break;
6   }
7   num++;
8 }
```

The terminal at the bottom shows the command executed and the output:

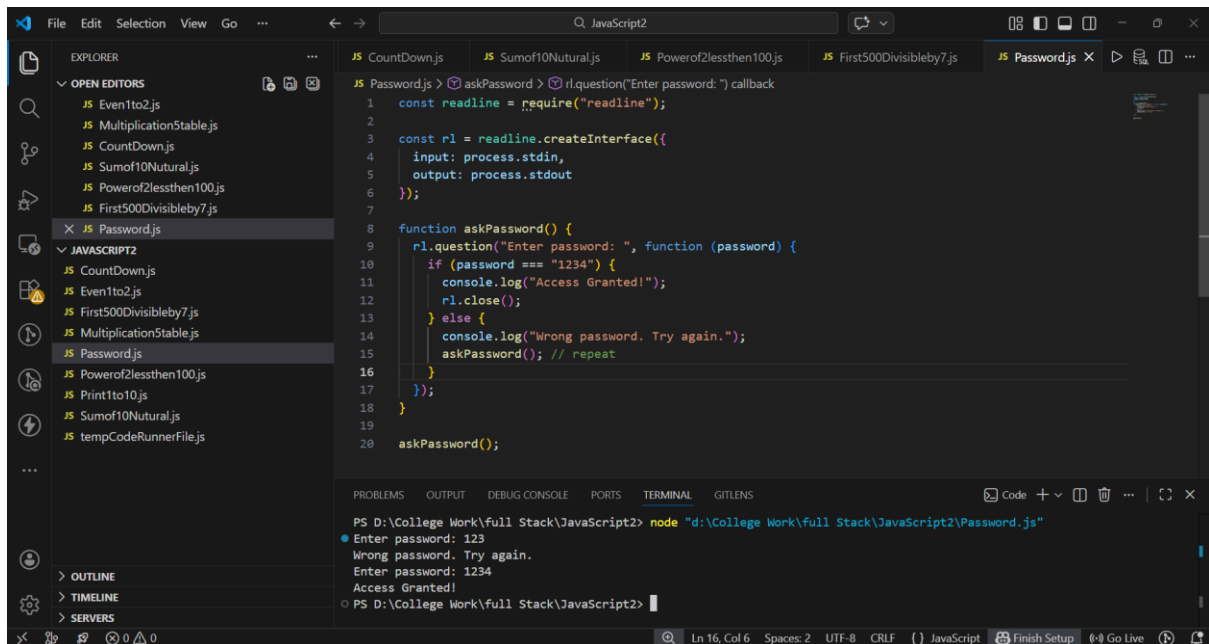
```
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\First500Divisibleby7.js"
First Number : 504
PS D:\College Work\full Stack\JavaScript2>
```

The Explorer sidebar on the left shows the project structure with the following files:

- EXPLORER
 - OPEN EDITORS
 - JS Even1to2.js
 - JS Multiplication5table.js
 - JS Countdown.js
 - JS Sumof10Natural.js
 - JS Powerof2lessthen100.js
 - JS First500Divisibleby7.js
 - JAVASCRIPT2
 - JS Countdown.js
 - JS Even1to2.js
 - JS First500Divisibleby7.js
 - JS Multiplication5table.js
 - JS Powerof2lessthen100.js
 - JS Print1to10.js
 - JS Sumof10Natural.js
 - JS tempCodeRunnerFile.js

🚦 Do While Loop Tasks :

1. Create a loop that "asks" for a password and keeps running until the password equals "1234"



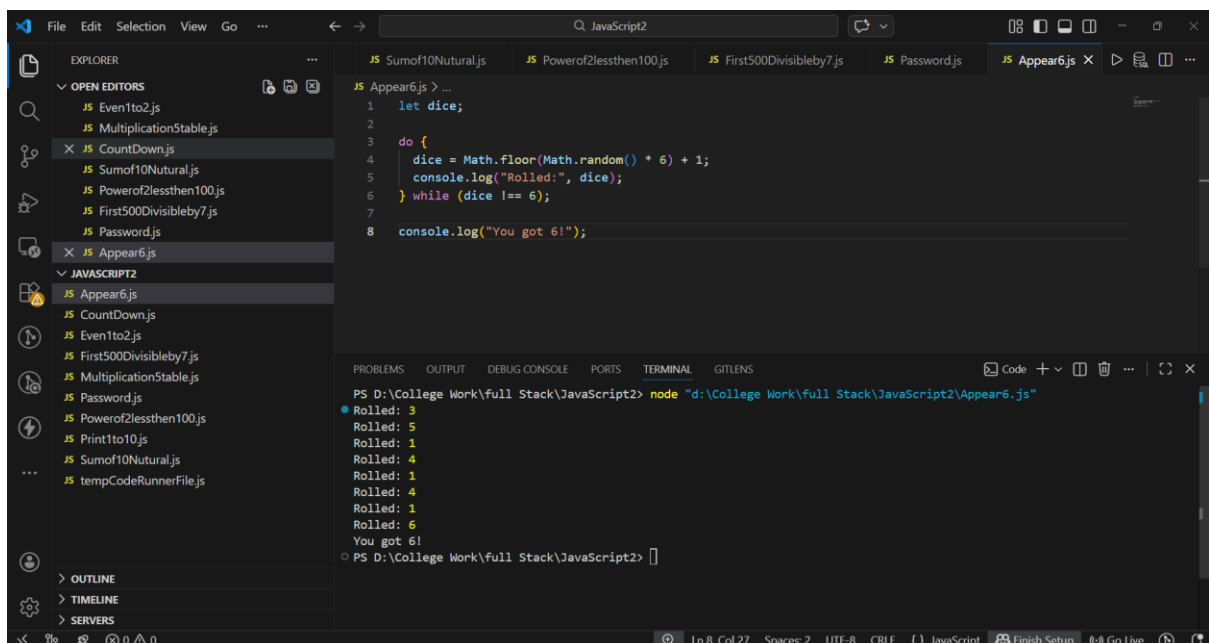
```
JS Password.js > askPassword > rl.question("Enter password: ") callback
1 const readline = require("readline");
2
3 const rl = readline.createInterface({
4   input: process.stdin,
5   output: process.stdout
6 });
7
8 function askPassword() {
9   rl.question("Enter password: ", function (password) {
10     if (password === "1234") {
11       console.log("Access Granted!");
12       rl.close();
13     } else {
14       console.log("Wrong password. Try again.");
15       askPassword(); // repeat
16     }
17   });
18 }
19
20 askPassword();
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL GITLENS

PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Password.js"

Enter password: 123
Wrong password. Try again.
Enter password: 1234
Access Granted!
PS D:\College Work\full Stack\JavaScript2>

2. Simulate rolling a 6-sided die. Keep rolling and printing the result until you hit a 6



```
JS Sumof10Natural.js JS Powerof2lessthen100.js JS First500Divisibleby7.js JS Password.js JS Appear6.js
1 let dice;
2
3 do {
4   dice = Math.floor(Math.random() * 6) + 1;
5   console.log("Rolled:", dice);
6 } while (dice !== 6);
7
8 console.log("You got 6!");
```

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL GITLENS

PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Appear6.js"

Rolled: 3
Rolled: 5
Rolled: 1
Rolled: 4
Rolled: 1
Rolled: 4
Rolled: 1
Rolled: 6
You got 6!
PS D:\College Work\full Stack\JavaScript2>

3. Calculate the factorial of a number using a do...while loop.

```
File Edit Selection View Go ... JavaScript2
EXPLORER
  OPEN EDITORS
    Even1to2.js
    Multiplication5table.js
    Countdown.js
    Sumof10Natural.js
    Powerof2lessthen100.js
    First500Divisibleby7.js
    Password.js
    Appear6.js
    Factorial.js
  JAVASCRIPT2
    Appear6.js
    Countdown.js
    Even1to2.js
    Factorial.js
    First500Divisibleby7.js
    Multiplication5table.js
    Password.js
    Powerof2lessthen100.js
    Print1to10.js
    Sumof10Natural.js
    tempCodeRunnerFile.js
  OUTLINE
  TIMELINE
  SERVERS

Factorial.js
1 let num = 5
2 let i = 1
3 let factorial = 1
4 do {
5   factorial *= i;
6   i++;
7 } while (i <= num)
8 console.log('Factorial =', factorial)

PROBLEMS OUTPUT DEBUG CONSOLE PORTS TERMINAL GITLENS
PS D:\College Work\full Stack\JavaScript2> node "d:\College Work\full Stack\JavaScript2\Factorial.js"
Factorial = 120
PS D:\College Work\full Stack\JavaScript2>
```